

Motivation

Stationarity is a fundamental assumption in time series analysis, however many real-world processes exhibit abrupt changes in their underlying distribution. A change point is a discontinuity where the time series changes in parameter (e.g. mean, variance) or in distribution. Identifying such points enables us to treat the series as a sequence of piecewise stationary segments and ensure accurate analysis by preserving the validity of the underlying stationarity assumption. This poster focuses on changes in mean in a discrete time series $\{X_t\}_{t=1}^n$, which we assume can be modelled as an underlying signal plus an additive noise term [1]:

$$X_t = f(t) + \varepsilon_t,$$

where $f(t)$ represents the underlying signal and $\varepsilon_t \stackrel{i.i.d}{\sim} \mathcal{N}(0, \sigma^2)$.

Single Change Point Model

We can specify a single change point at index τ by writing:

$$X_t = \begin{cases} f_1(t) + \varepsilon_t & \text{for } 1 \leq t \leq \tau, \\ f_2(t) + \varepsilon_t & \text{for } \tau + 1 \leq t \leq n, \end{cases}$$

such that $f_1(t) \neq f_2(t)$. For a single change in mean we have $\mu_1 \neq \mu_2$ and:

$$X_t = \begin{cases} \mu_1 + \varepsilon_t & \text{for } 1 \leq t \leq \tau, \\ \mu_2 + \varepsilon_t & \text{for } \tau + 1 \leq t \leq n. \end{cases} \iff X_t \sim \begin{cases} \mathcal{N}(\mu_1, \sigma^2) & \text{for } 1 \leq t \leq \tau, \\ \mathcal{N}(\mu_2, \sigma^2) & \text{for } \tau + 1 \leq t \leq n. \end{cases}$$

Testing for a Single Change

We want to test the following hypotheses:

$$\mathcal{H}_0 : \theta = \theta_0 \forall 1 \leq t \leq n, \quad \mathcal{H}_1 : \begin{cases} \theta = \theta_1 & \text{for } 1 \leq t \leq \tau, \\ \theta = \theta_2 & \text{for } \tau + 1 \leq t \leq n, \end{cases}$$

such that $\theta_1 \neq \theta_2$. One way to do this is by constructing a test statistic comparing the likelihood of $\mathcal{H}_0$ with that of $\mathcal{H}_1$ given the data.

Definition (Generalised Log Likelihood Ratio Test Statistic):

The GLLR test statistic at index t is given by:

$$\mathcal{LR}_t = 2 \left[\mathcal{L}(\hat{\theta}_1; X_{1:t}) + \mathcal{L}(\hat{\theta}_2; X_{(t+1):n}) - \mathcal{L}(\hat{\theta}_0; X_{1:n}) \right],$$

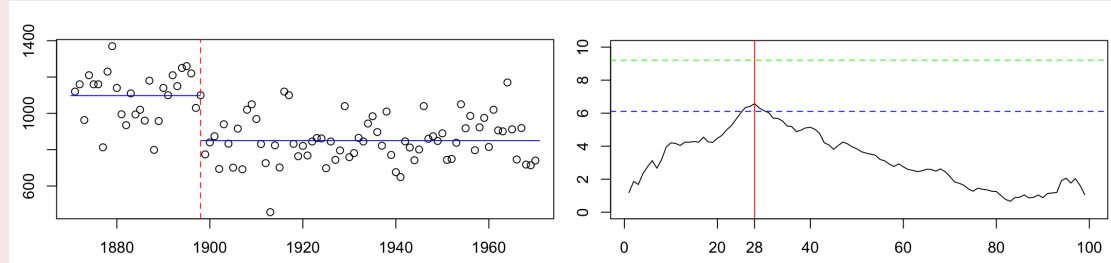
where $\hat{\theta}_i$ represents the maximum likelihood estimator for θ_i . For a change in mean problem, we have:

$$\mathcal{LR}_t = \frac{1}{\sigma^2} \left[\sum_{t=1}^n (X_t - \bar{X}_{1:n})^2 - \sum_{t=1}^{\tau} (X_t - \bar{X}_{1:\tau})^2 - \sum_{t=(\tau+1)}^n (X_t - \bar{X}_{(\tau+1):n})^2 \right].$$

To test for a single change, we compute $\mathcal{LR}_t$ for each $t \in \{1, \dots, n-1\}$. Our estimator for τ is then $\hat{\tau} = \arg \max_{\{t=1, \dots, n-1\}} \mathcal{LR}_t$, representing the index most likely to be a change point [1].

Nile River Data

This example demonstrates the GLLR test applied to the Nile dataset from the R *datasets* package [2]. It identifies a change point at $t = 28$, corresponding to the year 1898, during which a dam was built. This indicates the test detected a “true” change.



(a) Volume of Nile at Aswan between 1871 and 1970 (black), with identified changepoint at $\hat{\tau} = 1898$ (red dotted) and segment means (blue). (b) GLLR statistic for each index of Nile data, maximised at $t = 28$ (red). Two thresholds HQIC ≈ 6.11 (blue) and BIC ≈ 9.21 (green).

Figure 1. Application of single change point test using GLLR test statistic on Nile data ($n = 100$) [2].

Choice of Threshold

Once we have computed our maximum test statistic, $\mathcal{LR}_{\hat{\tau}} = \max_{\{t=1, \dots, n-1\}} \mathcal{LR}_t$, we must determine whether or not this corresponds to a change. We declare a change if $\mathcal{LR}_{\hat{\tau}} > c$, for some threshold c .

- The **Hannan-Quinn Information Criterion (HQIC)** gives threshold $c = 4 \log \log n$ [3]. It is derived from the Law of the Iterated Logarithm which states that extreme maximum fluctuations of the test statistic are asymptotically bounded by $2 \log \log n$ under $\mathcal{H}_0$. As we are estimating 2 parameters, both the location of the change and the parameter, we scale this to $4 \log \log n$.
- The **Bayes Information Criterion (BIC)** gives threshold $c = 2 \log n$, and is derived from Bayesian posterior probabilities [4]. The BIC is larger and so more conservative than the HQIC, which can result in false negatives if the change is small relative to the noise. Conversely, the lighter penalty of the HQIC threshold can increase the risk of false positives in noisy data, particularly if the underlying data deviates from the normal distribution.

For example, in fig. 1b, the change surpasses the HQIC threshold but not the BIC threshold, as it is a small change relative to the noise. Both thresholds are strongly consistent, meaning that the probability that they select the true model converges to 1 as $n \rightarrow \infty$.

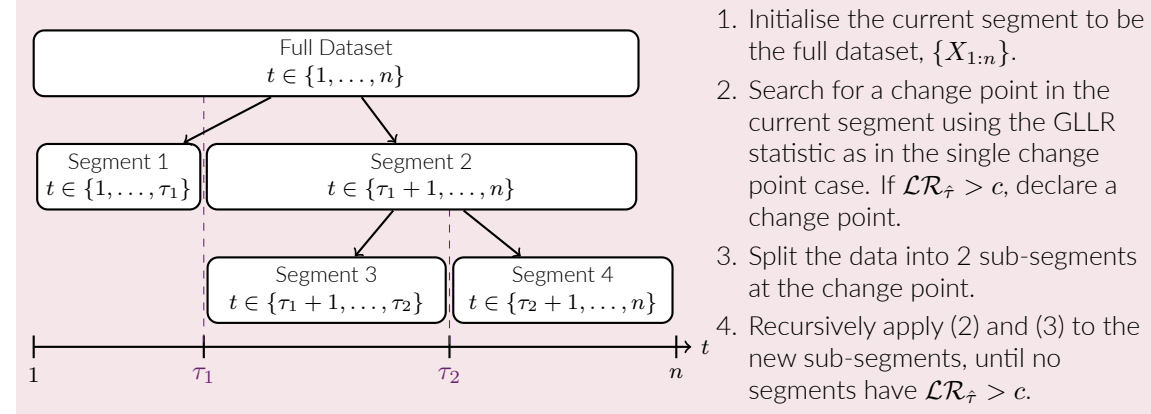
Detecting Multiple Change Points

We now consider the case of multiple changes in mean. We assume there are k change points at indices $\tau_1, \dots, \tau_k$, and that our underlying signal is $f_i(t) = \mu_i$, for $1 \leq i \leq k+1$. Then we have:

$$X_t \sim \mathcal{N}(\mu_i, \sigma), \quad \text{for } \tau_{i-1} < t \leq \tau_i,$$

with the convention that $\tau_0 = 1$ and $\tau_{k+1} = n$.

Top-Down Algorithm: Binary Segmentation



Binary segmentation has complexity $\mathbf{O(n \log n)}$, since it requires $O(n)$ operations per split across an expected tree depth of $O(\log n)$ [5]. While it is suitable for identifying **major global trends**, its **greedy** nature is susceptible to missing shorter changes that occur close together as they ‘mask’ one another in the global view. For this reason, a more lenient threshold such as the HQIC threshold is often necessary to avoid **early termination**.

The initial global split is typically the most unreliable, since the model assumes a single change point and so multiple true change points are not modelled. This artificially inflates the global variance estimate, reducing the GLLR statistic. To mitigate this masking effect, **Wild Binary Segmentation** uses random sub-intervals, ensuring there is no step which tests using the entire dataset at once [6].

Bottom-Up Algorithm

1. Initialise each point $t \in \{1, \dots, n\}$ as its own segment.
2. Calculate the GLLR test statistic for each adjacent pair of segments.
3. Find the adjacent pair with the smallest GLLR test statistic. If this statistic is lower than the threshold, merge them into a single segment.
4. Update the test statistics to account for the new segment structure.
5. Repeat (3) and (4) until the smallest GLLR test statistic across all segment pairs is greater than the threshold.

The bottom-up algorithm has complexity $\mathbf{O(n \log n)}$ when optimised to use a min-heap to store the test statistic [7]. This is because we expect to perform n merges, each of which takes $O(\log n)$ complexity to update the test statistics adjacent to the merge point. Unlike binary segmentation, which suffers from a global masking effect, bottom-up considers **local test statistics**, making it highly effective in **detecting shorter changes**. A disadvantage of this is its tendency to **over-segment** the data, mistaking noise for an underlying change. To prevent this, we could initialise with larger segments or specify a minimum segment length.

Sliding Window Algorithm

1. For a given window size $w = 2h$, consider the first w observations.
2. Calculate the GLLR test statistic for a change at the midpoint of this window. If it is greater than the threshold, add the midpoint index and its statistic to a list of candidate change points.
3. Move the window forward by one observation and repeat (2) and (3) until the window reaches the end of the dataset.
4. Sort the candidate list in descending order based on GLLR test statistic. Declare a change at the index with the highest statistic, and remove other candidate indices within η observations of the change point. Repeat this process until the list of candidates is empty.

The sliding window algorithm is fast, with complexity $\mathbf{O(n)}$, as it only requires a single pass of the data [7]. Note that this assumes $j \ll n$, since the pruning step (5) has complexity $O(j \log j)$, for j candidate points. While bottom-up segments can be imbalanced, the sliding window has a **fixed width**, and tests for the change point at its **midpoint**, which maximises statistical power. The main disadvantage of the sliding window algorithm is that its accuracy **highly depends on the choice of w** . If w is too large, the algorithm mirrors binary segmentation, finding large global changes only. Conversely, if w is too small, the algorithm can mistake noise for a change, much like in the bottom-up case. The **Multiscale Sliding Window** algorithm mitigates this by simultaneously testing with multiple window sizes, enabling detection of changes of varying scales [8].

UK Driver Deaths Data

The three algorithms are applied to the ‘UK Driver Deaths’ dataset from R [2], which details the number of drivers killed or seriously injured in the UK over a period encompassing two changes:

- **1972:** Seatbelts compulsory in all new cars.
- **1983** Seatbelts legally required to be worn.

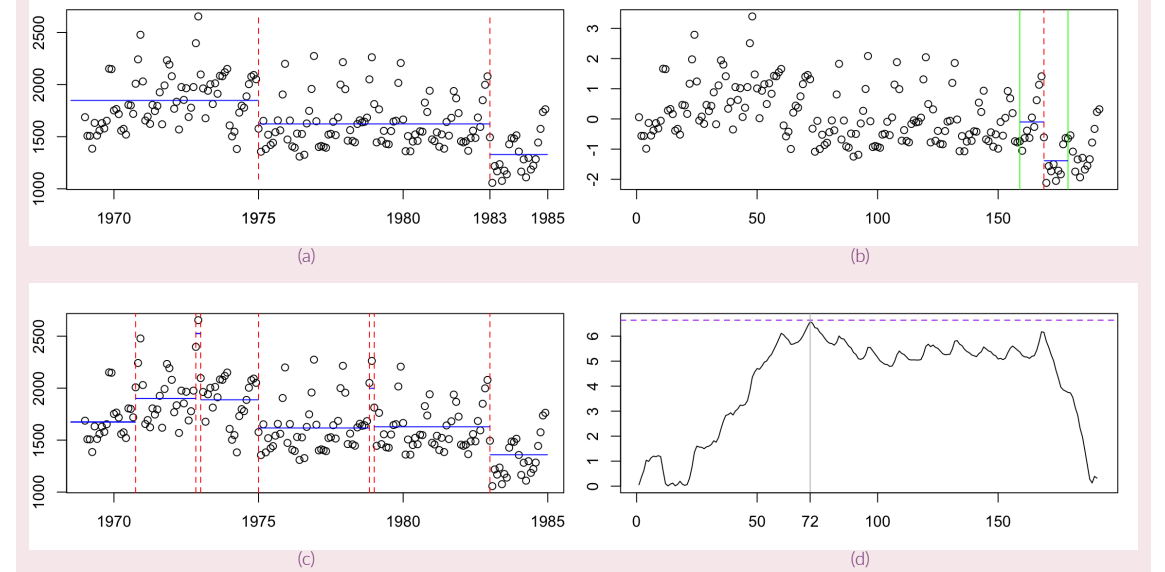


Figure 2. Application of multiple change point detection algorithms to the ‘UK Driver Deaths’ data [2] with HQIC threshold $c = 4 \log \log 192$. Raw data (black circles), detected change points (red dashed lines) and segment means (blue solid lines). (a) Final results of sliding window algorithm with $h = 10, \eta = 5$. (b) Intermediate step of sliding window with test window 158 to 178 (green) and detected change at $\hat{\tau} = 168$ corresponding to the change in 1983. (c) Results of implementing bottom up with HQIC threshold. (d) GLLR statistic for first step of binary segmentation. Maximum statistic at $\hat{\tau} = 72$ fails to exceed HQIC threshold (dashed purple), leading to early termination.

The sliding window results (fig. 2a) were obtained using the HQIC threshold. When the bottom-up algorithm was implemented with the same threshold, the data was estimated to have 7 changes, misclassifying noise as underlying changes (fig. 2c). Increasing the threshold to $2 \log n$ (BIC) identified the same change points as the sliding window algorithm (fig. 2a), which align with the underlying legislative changes. Conversely, implementing binary segmentation with the HQIC threshold led to termination at the first step due to masking (fig. 2d). Using a lower threshold of $2 \log \log n$ resulted in finding changes aligning with the legislative changes, with an additional change at $t = 10$ (possibly a false positive due to low threshold). This highlights the importance of threshold selection and demonstrates the contrasting methodologies: binary segmentation requires sufficient evidence to split the data, whereas the bottom-up algorithm requires sufficient evidence to merge. All three algorithms, after tuning the threshold, detected change points that reflect the real-world dynamics. The 1972 legislation produced a delayed change in the time series, whereas the 1983 law resulted in a sharp, instantaneous change.

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Poster based on template from <https://github.com/andiac/gemini-cam>